FIITJEE

# JEE (Advanced)-2018

ALL INDIA TEST SERIES

# CONCEPT RECAPITULATION TEST - IV

## Paper 2

**Time Allotted: 3 Hours** **Maximum Marks: 231**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

## INSTRUCTIONS

### A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into three sections: **Section-A, Section-C & Section-D**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

### B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

### C. Marking Scheme For All Three Parts.

1. ***Section-A (01 – 03, 24 – 26, 47 – 49)*** *contains 9 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*
   ***Section-A (04 – 08, 27 – 31, 50 – 54)*** *contains 15 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for correct answer and* ***–2 marks*** *for wrong answer.*
   Partial Marks **+1** for each correct option provided no incorrect options is selected.
   ***Section-A (09 – 10, 32 – 33, 55 – 56)*** *contains 3 paragraphs. Based upon paragraph, 2 multiple choice questions have to be answered. Each question has* ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer. There is no negative marking.*
2. ***Section-C (11 – 20, 34 – 43, 57 – 66)*** *contains 30 Numerical based questions with answer as numerical value from* ***0 to 9*** *and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*
3. ***Section-D (21 – 23, 44 – 46, 67 – 69)*** *contains 9 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

| | |
|---|---|
| **Name of the Candidate** | |
| **Enrolment No.** | |

## Useful Data

### PHYSICS

| | |
|---|---|
| Acceleration due to gravity | $g = 10\ m/s^2$ |
| Planck constant | $h = 6.6 \times 10^{-34}$ J-s |
| Charge of electron | $e = 1.6 \times 10^{-19}$ C |
| Mass of electron | $m_e = 9.1 \times 10^{-31}$ kg |
| Permittivity of free space | $\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$ |
| Density of water | $\rho_{water} = 10^3\ kg/m^3$ |
| Atmospheric pressure | $P_a = 10^5\ N/m^2$ |
| Gas constant | $R = 8.314\ J\ K^{-1}\ mol^{-1}$ |

### CHEMISTRY

| | | | |
|---|---|---|---|
| Gas Constant | R | = | $8.314\ J\ K^{-1}\ mol^{-1}$ |
| | | = | $0.0821$ Lit atm $K^{-1}\ mol^{-1}$ |
| | | = | $1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$ |
| Avogadro's Number | $N_a$ | = | $6.023 \times 10^{23}$ |
| Planck's constant | h | = | $6.625 \times 10^{-34}$ J·s |
| | | = | $6.625 \times 10^{-27}$ erg·s |
| 1 Faraday | | = | 96500 coulomb |
| 1 calorie | | = | 4.2 joule |
| 1 amu | | = | $1.66 \times 10^{-27}$ kg |
| 1 eV | | = | $1.6 \times 10^{-19}$ J |

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

**PART – I (Physics), PART – II (Chemistry), PART – III (Mathematics):**
**(SECTION – D)**

For questions **21 to 23, 44 to 46, 67 to 69.**
Numerical answer type questions with answer XXXXX. XX

If answer is 348.4 / 251.37 / 213

**Correct Method :**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 3 | 4 | 8 | . | 4 | 0 |
| 0 | 0 | 2 | 5 | 1 | . | 3 | 7 |
| 0 | 0 | 2 | 1 | 3 | . | 0 | 0 |

**Wrong Method :**

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 3 | 4 | 8 | | . | 4 | |
| 3 | 4 | 8 | | | . | | 4 |
| | | 3 | 4 | 8 | . | | 4 |
| | 3 | | 4 | 8 | . | 4 | |
| | 2 | | 5 | 1 | . | 3 | 7 |
| | | 2 | 1 | 3 | . | | |
| | | 2 | 1 | 3 | . | 0 | |
| | | 2 | 1 | 3 | . | | 0 |
| | | 3 | 4 | 8 | . | 4 | 0 |
| | | 2 | 5 | 1 | . | 3 | 7 |
| | | 2 | 1 | 3 | . | 0 | 0 |

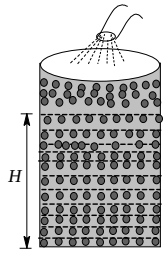
# Physics

## PART – I

### SECTION – A
### (One Options Correct Type)

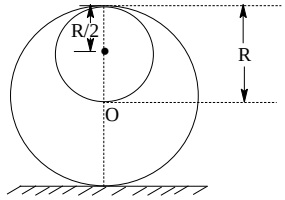
This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

1. A cylindrical container having radius r = 0.3 m has perforated wall. There is large number of uniformly spread small holes on the vertical wall occupying a fraction $\eta = 0.02$ of the entire area of the wall. To container, water is being fed to it at a constant rate $Q(m^3s^{-1})$ and at steady state its height becomes 2 m. Find $Q$.




(A) $0.32$ (B) $0.36$
(C) $0.2$ (D) $0.12$

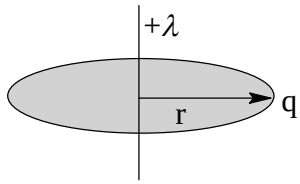
2. A spherical cavity of radius $\frac{R}{2}$ is removed from a solid sphere of radius R as shown in fig. The sphere is placed on a rough horizontal surface as shown. The sphere is given a gentle push. The sphere performs SHM and its time period is.




(A) $2\pi\sqrt{\frac{177R}{10g}}$ (B) $2\pi\sqrt{\frac{77R}{10g}}$
(C) $2\pi\sqrt{\frac{7R}{10g}}$ (D) $2\pi\sqrt{\frac{177R}{g}}$

***Space for Rough work***

3. A particle of charge -q and mass m moves in a circle of radius r around an infinitely long charge of linear charge density $+\lambda$ .Then time period will be



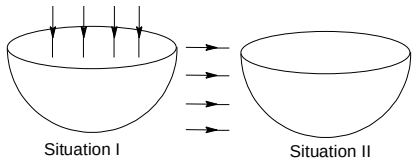

(A) $T=2\pi r\sqrt{\frac{m}{2k\lambda q}}$ (B) $T=2\pi r\frac{m}{k\lambda q}$

(C) $T=\frac{1}{2\pi r}\sqrt{\frac{2k\lambda q}{m}}$ (D) $T=\frac{1}{2\pi r}\sqrt{\frac{m}{2k\lambda q}}$

## (One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

4. In the figure a hemispherical bowl of radius R is shown. Electric field of intensity E is present perpendicular to the circular cross section of the hemisphere.



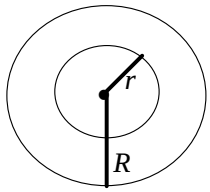

(A)The magnitude of electric flux through the hemispherical surface in situation I will be $E\pi R^2$

(B)The magnitude of electric flux through the hemispherical surface in situation I will be $2E\pi R^2$

(C)The magnitude of electric flux through the hemispherical surface in situation II will be $\frac{E\pi R^2}{2}$

(D)The magnitude of electric flux through the hemispherical surface in situation II will be zero

5. A soap bubble of radius r is formed inside another soap bubble of radius R(>r). The atmosphere pressure is $P_0$ and surface tension of the soap solution is T. Now the bigger bubble bursts. Assume that the excess pressure inside a bubble is small compared to $P_0$. For the smaller bubble choose the correct options:

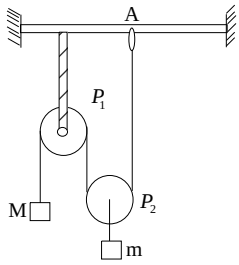



(A) The bubble shrinks (B) The bubble expands

(C) Change in its radius is $\frac{4Tr}{3P_0R}$ (D) Change in its radius is $\frac{3Tr}{4P_0R}$

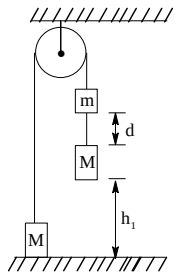
***Space for Rough work***

6. The arrangement shown in figure is in equilibrium with all strings vertical. The end A of the string is tied to a ring which can be slid slowly on the horizontal rod. Pulley $P_1$ is rigidly fixed but $P_2$ can move freely. A mass m is attached to the centre of pulley $P_2$ through a thread. Pulleys and strings are mass less.




(A) Block of mass M will move up as A is moved slowly to the right
(B) Block of mass m will move up as A is moved slowly to the right
(C) It is possible, for a particle position of A, that M has no acceleration but m does have acceleration
(D) The block of mass m will have a horizontal displacement

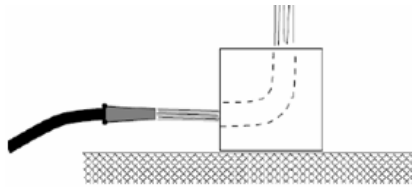
7. There are two weights of the same mass of 2.0 kg each attached to a string looped over a braked pulley. The left weight is lying on the floor, the right one is suspended 1.0 m above the floor. There is another weight of mass 0.5 kg attached to the string 20 cm above the right weight. The string except the part around the pulley is stretched vertically. We eventually release the pulley.




(A) The speed at which the lower right weight hits the floor is 1.48 m/s
(B) The height that the left weight rises to after the lower right weight falls inelastically to the floor is 1.185 m
(C) The maximum height that the lower right weight rises back to is 1.185 m
(D) The maximum height that the lower right weight rises back to is 0.31 m

***Space for Rough work***

8. A block of mass M, at rest on a frictionless surface, contains a curved channel. The channel is designed so that when a horizontal jet of fluid is aimed at it, the fluid is redirected 90 and shoots straight up and out of the top of the block. The flow of the jet is Q kg/s and has a velocity $v$ m/s. Assuming no gravity and ignoring any viscous effects. [Density of fluid is $\rho$ ]



(A) Speed of the block as a function of time is $v_b = v\left(1 - e^{-\frac{\rho Q t}{M}}\right)$

(B) Speed of the block as a function of time is $v_b = v\left(\frac{\rho Q t}{\rho Q t + M}\right)$

(C) Power delivered by the fluid to the block at $t = 0$ is zero

(D) Power delivered by the fluid to the block at $t = 0$ is $\rho Q v$

## (Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

### Paragraph for Question Nos. 9 and 10

In a homogeneous, non-magnetic, highly insulating and viscous medium, a moving particle experiences a viscous drag given by the law $\vec{f} = -b\vec{v}$ . Here b is a positive constant. A particle having charge q projected with an unknown velocity from a point in the medium almost stops (its speed becomes practically negligible) after travelling a distance of 10 m in a straight line. Now a uniform magnetic field is established in the region and the same particle is again launched with the same projection velocity directed perpendicular to the magnetic field.

9. In the presence of above magnetic field, if the particle almost stops at a point 6m away from the point projection the magnetic field is equal to:

(A) $\frac{4b}{3q}$ (B) $\frac{3b}{4q}$

(C) $\frac{4q}{3b}$ (D) $\frac{3q}{4b}$

***Space for Rough work***

10. The strength of the magnetic field is now doubled in magnitude. How far (in meters) from the point of projection will the particle come to an almost rest?

(A) $\frac{\sqrt{73}}{2}$ (B) $\frac{20}{\sqrt{73}}$

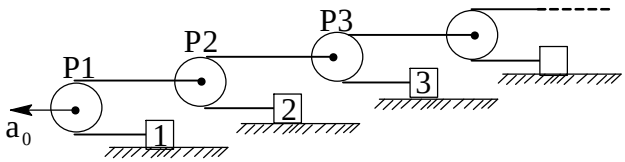
(C) $\frac{40}{\sqrt{73}}$ (D) $\frac{30}{\sqrt{73}}$

## SECTION – C
## (Single digit integer type)

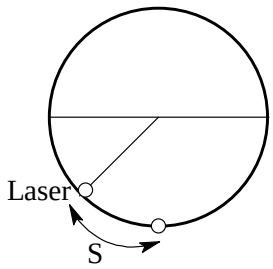
This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

11. A spherical planet of radius r, painted black, travels around the sun in a circular orbit at a distance D from its centre $(r<<<D)$. The sun, a sphere of radius R, radiates as a blackbody at a temperature $T_0=6000K$ and subtends an angle 0.01 rad as seen from the planet. The equilibrium temperature T of the planet is $x\times100$. Find x

12. In the figure all pulleys (P1, P2, P3…….) are massless and all the blocks (1, 2, 3…..) are identical, each having mass m. The system consists of infinite number of pulleys and blocks. Strings are light and in extension and horizontal surfaces are smooth. Pulley P1 is moved to left with a constant acceleration of $a_0$. The acceleration of block 1 will be (Assume the strings to remain horizontal) $ka_0$. Find 2k

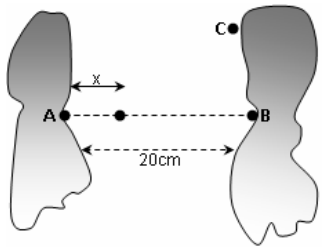



13. A horizontal cylindrical tank $6\pi m$ in diameter is half full of water. The space above the water is filled with a pressurized gas of unknown refractive index. A small laser can move all the curved bottom of the water and aims a light beam towards the centre of the water surface. When the laser has moved a distance $S=1m$ or more (measured along the curved surface) from the lowest point in the water, no light enters the gas. If the refractive index of the gas is $(\mu_{max}=4/3)$ $2/\sqrt{n}$ ,then n is

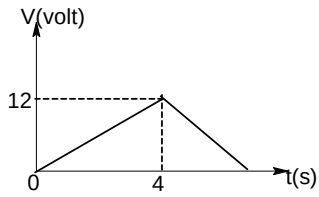



***Space for Rough work***

14. Two metallic bodies separated by a distance 20cm are given equal and opposite charges of magnitude $0.88\mu C$. The component of electric field along the line AB, between the plates, varies as $E_x = 3x^2 + 0.4 N/C$, where x (in meters) is the distance from one body towards the other as shown. The capacitance of the system is $n \times 10\mu F$. Find n




15. A pure inductor of inductance 3H is given across which a potential difference varying with time is shown in the adjoining figure. At t=0, current in the inductor is zero. If the current in the circuit at t=3s is N amperes. Find N



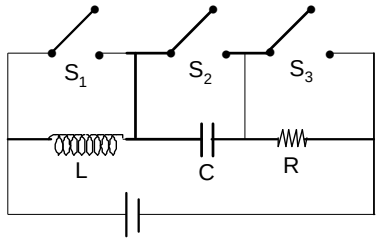

16. Nuclei of radioactive element A are produced at rate $'t^2'$ at any time t. The element A has decay constant $\lambda$. Let N be the number of nuclei of element A at anytime t. AT time $t=t_0$, $\frac{dN}{dt}$, is minimum. Then the number of nuclei of element A at time $t=t_0$ is found to be $\frac{Nt_0 - \lambda t_0^2}{\lambda}$. Find the value of N

17. If 10000 V is applied across an X-ray tube, if the ratio of de-Broglie wavelength of the incident electrons to the shortest wavelength of X-ray produced (e/m for electron is $1.8 \times 10^{11} C\,kg^{-1}$) is r. Find 2/5r.

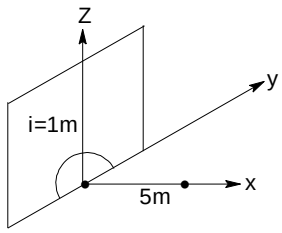
18. The internal energy of a monatomic ideal gas is 1.5 nRT. One mole of helium is kept in a cylinder of cross-section 8.5 $cm^2$. The cylinder is closed by a light frictionless piston. The gas is heated slowly in a process during which a total of 42 J heat is given to the gas. The temperature rises through 2°C. The distance moved by the piston is given as $a \times 10^b$ m in scientific notation. Find the value of a + b. [Take R=25/3 in SI unit, atmospheric pressure = 100 kPa]

***Space for Rough work***

19. Consider the circuit shown in figure. With switch $S_1$ closed and the other two switches open, the circuit has a time constant 0.05 sec. With switch $S_2$ closed and the other two switches open, the circuit has a time constant 2 sec. With switch $S_3$ closed and the other two switches open, the circuit oscillates with a period T. Find T (in sec). (Take $\pi^2$ = 10)




20. A plane mirror of semicircular shape (radius = 1m) is placed in y-z plane with its diameter along y-axis and centre at origin. An object is placed on the axis of mirror at point (5, 0, 0). At a distance of 10 m from plane mirror there is a wall also parallel to y-z plane. The ratio of area of spot of light on wall to that of mirror is

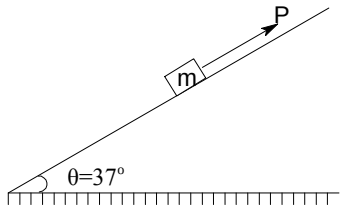



## SECTION – D
## (Numerical Based XXXXX.XX answer Type)

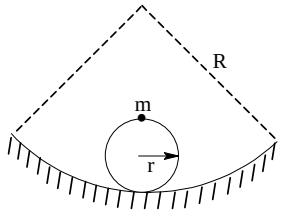
This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

21. Steel has density 7800 kg/m$^3$ and will break if tensile stress exceeds 7.0 × 10$^8$ N/m$^2$. A person wants to make a guitar string from 4.0 gm of this steel. The guitar string must be able to withstand a tension of 900 N. The maximum speed of transverse wave can be $\text{x.xx} \times 100$ m/s

22. A block of mass m is being pulled up the rough incline, inclined at an angle 37° with horizontal by an agent delivering constant power P. The coefficient of friction between the block and the incline is μ. Find the maximum speed (in km/hr) of the block during the course of ascent. Give the answer as x.xxx [Take : P = 60 W, m = 1 kg, μ = 0.5]




23. A ring of radius r and mass 2m, rests on a rough circular track of radius R.A small point mass of mass m is mounted on its periphery. If friction is sufficient to avoid any slipping then the minimum ratio of $\frac{r}{R}$ is x.xx




***Space for Rough work***

# Chemistry

## PART – II

### SECTION – A
### (One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

24. Give the correct order of initials T(True) or F(False) for following statements:
(I) In an antifluorite structure, anions form FCC lattice and cations occupy all tetrahedral voids
(II) If the radius of cations and anions are 0.2 and 0.95 then co-ordination number of cation in the crystal is 4
(III) An atom/ion is transferred from a lattice site to interstitial position in Frankel defect
(IV) Density of crystal always increase due to substitutional impurity defect
(A) TFFF (B) FTTF
(C) TFFT (D) TFTF

25. When one mole of A(g) is introduced in a closed rigid one litre vessel maintained at constant temperature, the following equilibria are established:

$A(g) \rightleftharpoons B(g) + C(g); \quad K_{C_1}$

$C(g) \rightleftharpoons D(g) + B(g); \quad K_{C_2}$

The pressure at equilibrium is twice the initial pressure if $\frac{[C]_{eq}}{[B]_{eq}} = \frac{1}{5}$, then $\frac{K_{c_2}}{K_{c_1}}$ is equal to

(A) 16/5 (B) 4
(C) 16 (D) 5/16

26. The specific conductivity of 0.5 M aq. solution of a monobasic acid (HA) after subtracting the specific conductance of water is found to be 0.006 $Scm^{-1}$. If its molar conductivity at infinite dilution is 200 $Scm^2mol^{-1}$, then the osmotic pressure of 0.1 M aq. solution of HA will be ($R = 0.0821\ dm^3\ atm\ mol^{-1}K^{-1}$)whole experiment took at 1365K.
(A) 0.72 atm (B) 1.72 atm
(C) 12.72 atm (D) 7.72 atm

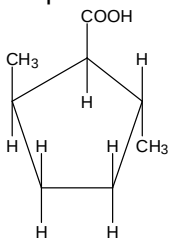
***Space for Rough work***

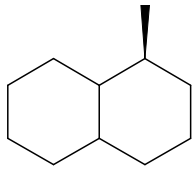
## (One or More than one correct type)

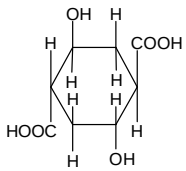
This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

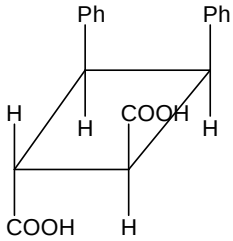
27. Copper glance to pure copper undergoes which of the following processes?
(A) Hydrometallurgy (B) Electrolytic refining
(C) Roasting (D) poling

28. Which species do not exhibit a plane of symmetry?

(A) 


(B) 

(C) 

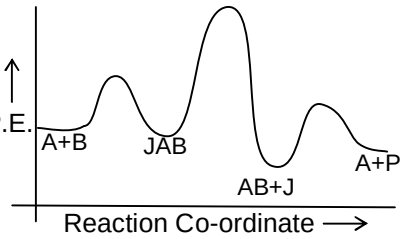

(D) 

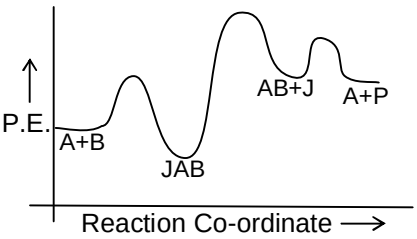

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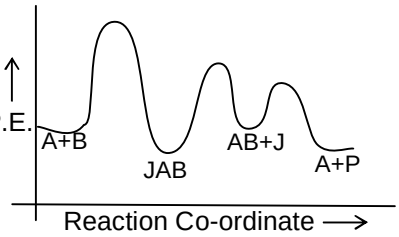
29. The following mechanism has been proposed for the exothermic catalysed complex reaction:

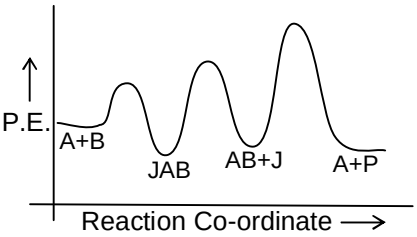
$$A + B \underset{}{\overset{\text{Fast}}{\rightleftharpoons}} JAB \xrightarrow{k_1} AB + J \xrightarrow{k_2} P + A$$

if $k_1$ is much smaller than $k_2$, then the incorrect qualitative plot of potential energy (PE) vs reaction coordinate for the above reaction is/are

(A) 


(B) 


(C) 


(D) 


30. Which of the following statements are correct?
(A) The crystal lattice of ice is mostly formed by covalent as well as hydrogen bonds
(B) The density of water increases when heated from $0^oC$ to $4^oC$ due to the change in the structure of cluster of water molecules
(C) Above $4^oC$, thermal agitation of water molecules increases and water starts expanding
(D) The density of water increases from $0^oC$ to a maximum at $4^oC$ and the entropy of the system increases

31. Elements of $15^{th}$ group form hydrides of general formula $MH_3$. Which of the following characteristic(s) of these hydrides decreases regularly on moving down the group (from $NH_3$ to $BiH_3$)?
(A) decomposition temperature
(B) boiling temperature
(C) reducing character
(D) HMH bond angle

***Space for Rough work***

## (Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

### Paragraph for Question Nos. 32 and 33

The concentration of potassium ions inside a biological cell is at least twenty times higher than the outside. The resulting potential difference across the cell is important in several processes such as transmission of nerve impulses and maintaining the ion balance. A simple model for such a concentration cell involving a metal M is:

$M(s) \mid M^+(aq; 0.05 \text{ molar}) \mid\mid M^+(aq), 1 \text{ molar}) \mid M(s)$

For the above electrolytic cell the magnitude of the cell potential $|E_{cell}| = 70$ mV.

32. For the above cell

(A) $E_{cell} < 0; \Delta G > 0$ (B) $E_{cell} > 0; \Delta G < 0$

(C) $E_{cell} < 0; \Delta G^o > 0$ (D) $E_{cell} > 0; \Delta G^o > 0$

33. If the 0.05 molar solution of $M^+$ is replaced by 0.0025 molar $M^+$ solution, then the magnitude of the cell potential would be

(A) 35 mV (B) 70 mV

(C) 140 mV (D) 700 mV

## SECTION – C
## (Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

34. How many is/are correct for Glycine.

(A) Forms salt with strong inorganic acids

(B) It reacts with acetyl chloride to give acetyl derivative.

(C) When it is treated with $HNO_2$, $N_2$ is evolved.

(D) With Nitrosyl chloride $(NOCl)$ it forms chloro acetic acid.

(E) It can be esterified by an alcohol in presence of inorganic acid.

(F) It can form coordinated compounds with metal ions.

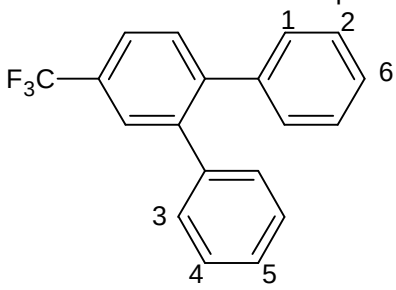
(G) Can be reduced to aminoalcohols by $LiAlH_4$.

(H) It is optically inactive.

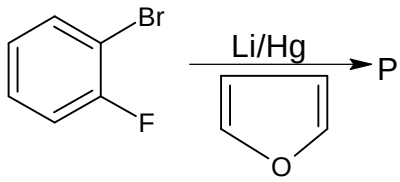
(I) Can be prepared by the action of concentrated ammonium hydroxide solution on chloroacetic acid

***Space for Rough work***

35. Which is the most favoured position for the attack of an electrophile




36.




What is the degree of unsaturation in P.

37. Calculate the value of $\frac{Z}{10}$. Where

Z = Co-ordination number of 2D-square close packing

+

Coordination number of 2D-hcp

+

Co-ordination number of 3D-square close packing

+

Co-ordination number of 3D, ABCABC…….Packing

+

Co-ordination number of 3D, ABAB……. packing

38. The standard reduction potential of a silver chloride electrode (metal-sparingly soluble salt electrode) is 0.209 V and for silver electrode is 0.80V. If the moles of $AgCl$ that can dissolve in 10L of a 0.01M NaCl solution is represented as $10^{-Z}$ then find the value of Z.

***Space for Rough work***

39. In the given radioactive disintegration series,
$^{232}_{90}\text{Th} \rightarrow ^{208}_{82}\text{Pb}$
Calculate value of (n+2)
Where value of n is number of isobars formed in this series, suppose there is successive emission of $\beta$-particles .

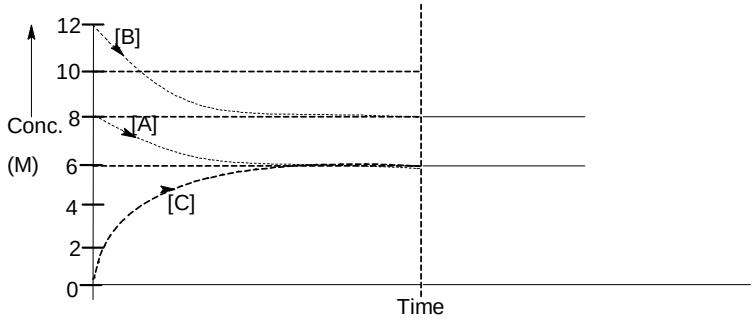
40. Among the following, the total number of compounds whose aqueous solution turns red litmus paper blue is:

| | | | | | |
|---|---|---|---|---|---|
| $NaCN$ | $KCl$ | $CH_3COONH_4$ | $NaH_2PO_4$ | $ZnCl_2$ | $Na_3PO_4$ |
| $Fe(NO_3)_3$ | $Na_2CO_3$ | $NH_4Cl$ | $NaHCO_3$ | $Na_2C_2O_4$ | $Na_2HPO_4$ |

**Given:**

| **Acid** | $Ka_1$ | $Ka_2$ | $Ka_3$ |
|---|---|---|---|
| $H_3PO_4$ | $10^{-3}$ | $10^{-8}$ | $10^{-12}$ |
| $H_2CO_3$ | $10^{-6}$ | $10^{-11}$ | - |
| $H_2C_2O_4$ | $10^{-2}$ | $10^{-5}$ | - |

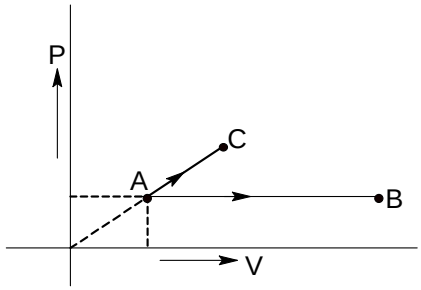
41. The gaseous reaction: $A(g)+nB(g) \rightleftharpoons mC(g)$ is represented by following curves




What is the value of $n+m$

***Space for Rough work***

42. One mole ideal monoatomic gas is heated gas is heated according to path AB and AC. If temperature of state B and C are equal. Calculate $\frac{q_{AC}}{q_{AB}} \times 10$.




43. How many of the following decompose easily as compared to $CaCO_3$ to evolve $CO_2$

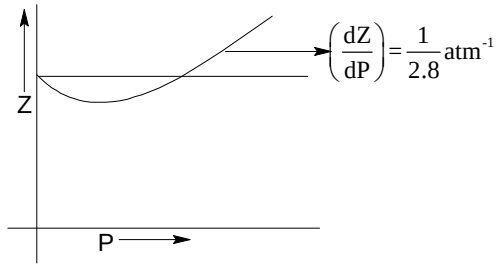
$CdCO_3$, $PdCO_3$, $COCO_3$, $MnCO_3$, $Na_2CO_3$, $BaCO_3$, $FeCO_3$, $BeCO_3$, $Ag_2CO_3$

## SECTION – D
## (Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

44. When a liquid that is immiscible with water was steam distilled at $95.2^oC$ at a total pressure of 748torr, the distillate contained 1.25g of the liquid per gram of water. The vapour pressure of water is 648 torr at $95.2^oC$, what is the molar mass of liquid in g/mol.

45. The graph of compressibility factor (Z) vs. P for one mole of a real gas is shown in following diagram. The graph is plotted at constant temperature 273K. If the slope of graph at very high pressure $\left(\frac{dZ}{dP}\right)$ is $\left(\frac{1}{2.8}\right) atm^{-1}$, then calculate volume of one mole of real gas molecules (in L/mol) Given:




$N_A = 6 \times 10^{23}$ and $R = \frac{22.4}{273} L\,atm\,K^{-1}\,mol^{-1}$

46. $40g\ Ba(MnO_4)_2\ (mol.mass = 375)$ sample containing some inert impurities in acidic medium completely reacts with 125mL of ''33.6 V'' of $H_2O_2$. What is the percentage purity of the sample?

***Space for Rough work***

# Mathematics

## PART – III

### SECTION – A
### (One Options Correct Type)

This section contains **3 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE** option is correct.

47. $I = \int \frac{x^2}{(x\cos x - \sin x)} \frac{dx}{(x\sin x + \cos x)}$

(A) $\ln\left|\frac{x\cos x - \sin x}{x\sin x + \cos x}\right| + c$ (B) $\ln\left|\frac{x\sin x + \cos x}{x\cos x - \sin x}\right| + c$

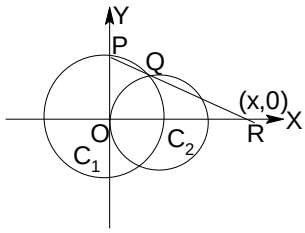
(C) $\frac{x\sin x + \cos x}{x\cos x - \sin x} + c$ (D) $\frac{x\cos x - \sin x}{x\sin x + \cos x} + c$

48. Find the minimum value of $(x_1 - x_2)^2 + \left(\sqrt{1 - x_1^2} - 12 + 2\sqrt{x_2}\right)^2$

(A) $\left(\sqrt{70} - 2\right)^2$ (B) $\left(\sqrt{80} - 1\right)^2$

(C) $\left(\sqrt{70} - 1\right)^2$ (D) $\left(\sqrt{80} + 1\right)^2$

49.




Find the limiting value of OR as $r \to 0^+$

$C_1 : x^2 + y^2 = r^2$

$C_2 : (x-1)^2 + y^2 = 1^2$

(A) 2 (B) 4

(C) 8 (D) 16

***Space for Rough work***

## (One or More than one correct type)

This section contains **FIVE** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

50. For the ellipse $12x^2 - 12xy + 7y^2 = 48$

(A)slope of major axis is $\frac{3}{2}$ and length =8

(B)slope of minor axis is $\frac{-2}{3}$ and length=6

(C)coordinate of foci are $(2,3)\text{ and }(-2,-3)$

(D)Equation of directrices are $2x+3y+14=0\text{ and }2x+3y-14=0$

51. Which of the following are correct

(A) If A and B are both symmetric or skew symmetric matrix and they are anticommutative and AB is skewsymmetric matrix

(B) $(\text{adjA})^{-1} = \frac{A}{|A|}$ (where A is any square matrix)

(C) For a square matrix A to be orthogonal it is necessary and sufficient that its $|A| = \pm 1$ and that of each element be equal to its cofactor taken with sign if $|A| = 1$ and with sign reversed if $|A| = -1$

(D) The product of 2 orthogonal matrices is orthogonal matrices is orthogonal and inverse of an orthogonal matrix is orthogonal.

52. If $A_1, A_2, A_3, .....A_{1006}$ be independent events such that $P(Ai) = \frac{1}{2i} (i=1,2,3,....1006)$ and probability that none of the events occurs be $\frac{\alpha!}{2^{\alpha}(\beta!)^2}$ then

(A) $\beta$ is of form $4k+2, k \in I$ (B) $\alpha = 2\beta$

(C) $\beta$ is a composite number (D) $\alpha$ is of form $4k, k \in I$

***Space for Rough work***

53. Consider the lines:

$L_1: \frac{x-2}{1} = \frac{y-1}{7} = \frac{z+2}{-5}$ $L_2: x-4=y+3=-z$

Then which of the following is/are correct?

(A) Point of entersection of $L_1$ and $L_2$ is (1,-6,3)

(B) Equation of plane containing $L_1$ and $L_2$ is $x+2y+3z+2=0$

(C) Acute angle between $L_1$ and $L_2$ is $\cos^{-1}\left(\frac{13}{15}\right)$

(D) Equation of plane containing $L_1$ and $L_2$ is $x+2y+3z+2=0$

54. Let $z_1, z_2$ and $z_3$ be three distinct complex numbers, satisfying $|z_1| = |z_2| = |z_3| = 1$, which of the following is/are correct.

(A) If $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$ then $\arg\left(\frac{z-z_1}{z-z_2}\right) > \frac{\pi}{4}$ where $|z| > 1$

(B) $|z_1z_2 + z_2z_3 + z_3z_1| = |z_1 + z_2 + z_3|$

(C) $\text{Im}\left(\frac{(z_1+z_2)(z_2+z_3)(z_3+z_1)}{z_1.z_2.z_3}\right) = 0$

(D) If $|z_1 - z_2| = \sqrt{2}|z_1 - z_3| = \sqrt{2}|z_2 - z_3|$, then $\text{Re}\left(\frac{z_3 - z_1}{z_3 - z_2}\right) = 0$

## (Paragraph Type)

This section contains **ONE** paragraph. Based on the paragraph, there are **TWO** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.

### Paragraph for Question Nos. 55 and 56

The centres of two circles $c_1$ and $c_2$ of radii 4 and 9 respectively are at a distance of 15units from each other. Let $L_1$ be one correct common tangent to these circles.

55. There are n circles which touches both the circle $C_1$ and $C_2$ and also the line $L_1$ then n is

(A) 2 (B) 4
(C) 3 (D) 8

56. Let $r_1, r_2, .....r_n$ be the radii of these circles then find $\max(r_1, r_2, .....r_n)$ ?

(A) $\frac{25}{2}$ (B) $\frac{50}{9}$
(C) 50 (D) 1

***Space for Rough work***

## SECTION – C
## (Single digit integer type)

This section contains **TEN** questions. The answer to each question is a single Digit integer ranging from 0 to 9, both inclusive.

57. Find the largest integer value of x satisfying the inequality $\log_{0.09}(x^2+2x) \geq \log_{0.3}\sqrt{x+2}$.

58. Let $H_n = 1+\frac{1}{2}+\frac{1}{3}+...+\frac{1}{n}$ and $T_n = \frac{1}{(n+1)H_nH_{n+1}}$, then the value of $(T_1+T_2+T_3+...+\infty)$ is:

59. Complex numbers a, b and c are zeroes of polynomials $P(z)=z^3+qz+r$ (where $q,r \in R$) and $|a^2|+|b^2|+|c^2|=250$. If the points corresponding to a, b and c in complex plane are the vertices of a right angled triangle with hypotenuse h, then $\left(\frac{h^2}{125}\right)$.

60. If N denotes the number of non-empty subsets of {1,2,3,…,12} having property that the sum of largest element and smallest element is 13, then number of divisors of N divisible by 5 is:

61. The number of distinct terms in expansion of $\left(x+y+z+\frac{1}{xy}+\frac{1}{yz}+\frac{1}{zx}\right)^2$ is m and that in expansion of $\left(x+y+z+\frac{1}{x}+\frac{1}{y}+\frac{1}{z}\right)^2$ is n, then $|m-n|$ is equal to:

***Space for Rough work***

62. There are 4 bowlers, 4 batsman and 1 all-rounder (who is bowler as well as batsman). Randomly a team of 4 players consisting of at least two bowlers and at least two batsman is formed. If the probability that all-rounder has been selected in team is $\frac{4}{p}$, then p is equal to:

63. Find the sum of all integral values of a values of a where $a \in [-10,10]$ such that the graph of the function $f(x) = \left|\left|x-2\right|-a\right|-3$ has exactly three x-intercepts.

64. Let $f : R \to R$ be a function defined as

$f(x) = \begin{cases} 1-|x|, & |x| \le 1 \\ 0, & |x| > 1 \end{cases}$ and $g(x) = f(x-1) + f(x+1) \forall x \in R$. Find number of points of non-differentiability of g(x) on R.

65. Suppose $\int \frac{1-7\cos^2 x}{\sin^7 x \cos^2 x} dx = \frac{g(x)}{\sin^7 x} + C$, where C is an arbitrary constant of integration. Then the value of $g'(0) + g''\left(\frac{\pi}{4}\right)$.

66. Let f be a differentiable function defined such that $f : [0,27] \to \left[\frac{1}{3}, 6\right]$ *and* $f'(x) < 0 \forall x \in D_f$ .If $\int_0^{27} x f'(x) dx = \lambda - 3\int_0^3 x^2 f(x^3) dx$ then find the value of $\lambda$.

**[Note:** $D_f$ denotes the domain of the function**.]**

***Space for Rough work***

## SECTION – D
## (Numerical Based XXXXX.XX answer Type)

This section contains **3 questions**. Each question, when worked out will result in numerical answer Type with answer xxxxx.xx.

67. The polynomial R(x) is the remainder upon dividing $x^{2007}$ by $x^2-5x+6$. If R (0) can be expressed as $ab(a^c-b^c)$, find the value of $\frac{a+b}{20}+c=?$

68. Let C be the set of 6 consonants {b,c,d,f,g,h} and V be a set of 5 vowels {a,e,I,o,u} and W be the set of seven letter words that can be formed with these 11 letters using both the following rules:
(a)The vowels can be consonants in the word must alternate.
(b)No letter can be used more than once in a single word.
If the number of words in the set W are K then find K/7.

69. Let f be real function defined on R (the set of real numbers) such that $f'(x)=100\ (x-1)(x-2)^2(x-3)^3.....(x-100)^{100}$, for all $x\in R$. If g is a function defined on R such that $\int_a^x e^{f(t)}dt=\int_0^x g(x-t)dt+2x+3$, If some of the all the values of x for which g(x) has a local extremum be $\lambda$ then find $\frac{\lambda}{3}$

***Space for Rough work***